

An Optimized Multiplicative Long-Short-Term-Memory Model for Bitcoin Price Prediction

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Abstract: Bitcoin is one of the first and the most popular cryptocurrency with a large market capitalization. The cryptocurrency market has been proven to be highly volatile due to various factors. Predicting the Bitcoin price is another complex task but still encourages investors to invest more to get more returns. Therefore, the need for Bitcoin price prediction arises to help traders and investigators predict accurate results nearer to the actual cryptocurrency prices. The importance of these prediction models is to have a significant economic ramification to pinpoint the purchase as well as sales. In this paper, the multiplicative Long Short Term Memory (mLSTM) Network model is used to predict the Bitcoin price optimized using the Cuckoo Search algorithm. Preliminary pre-processing steps are carried out to analyze the raw input data which helps in enhancing its quality, henceforth, the prediction accuracy. Then, the extraction of technical indicators is carried out using the least absolute shrinkage and selection operator (LASSO) and Fisher Score. The extracted features are fed as input to the mLSTM network, where auto-regression is used for error analysis. The outcomes are optimized with the Gamic Cuckoo Search Optimization approach (GCSO) where Cuckoo Search Optimization is integrated with Game Theory. To avoid overfitting or underfitting issues along with vanishing gradients a hybridization approach which is proposed using two industrial datasets and against several performance metrics approach is proposed and tested against several performance measures including

accuracy, root mean squared error (RMSE), mean absolute percentage error (MAPE),
mean absolute error (MAE), recall, and F1-score.

Keywords: Crypto-currency; Bitcoin; Price Prediction; Feature Extraction;
Multiplicative LSTM; Game Theory; Cuckoo Search Optimization

1. Introduction

For worldwide financial transactions, virtual currency is popularly used. The media and investors focus more on returns so cryptocurrencies are the crucial representatives, which may be attributed to the new features such as simplicity, transparency, and enhanced acceptance [1]. It is a type of currency that considers digital files as money and these files are created like cryptography (the science of information hiding) where digital signatures are utilized for making safe transactions and help people to check that the transactions are real. Presently, the most successful cryptocurrency bitcoin relies on the capitalization value of the cryptocurrency market. Presently, the market value of Bitcoin is 59.93 Trillion USD [2]. Bitcoin is a digital currency that operates freely over any centralized control or the oversight of government or banks. It relies on cryptography and peer-to-peer software. Here, a public ledger is used to record all the Bitcoin transactions, and copies are provided to the server all around the world. Additionally, the central bank or institution does not need Bitcoin for discharging and controlling it. The transactions are determined and preserved in the network with the help of cryptocurrency models in a peer-to-peer format for validating the transaction [3]. However, rather than using conventional credit cards, cryptocurrencies are faster, easier, and anonymous, benefiting payments [4].

Bitcoin is known as a popular cryptocurrency and is gaining popularity rapidly as a form of digital gold. The supply of bitcoin could boost its value significantly as it is volatile and considered a high-risk investment. Bitcoin has improved the acknowledgement in society because of its new features, more than 1,00,000 companies are interested in Bitcoin [5]. Different well-known companies such as CVS, Amazon, Expedia, Dell, Paypal, Home Depot, Target, Subway, Gap, Victoria's Secret, and a few more are interested in Bitcoin [6]. Further, the list of companies is growing continuously. Presently, authors of [7] represent other lists that have retail companies such as eGifter, Overstock, Microsoft (payment for movies, shopping, apps, and games), Newegg, and Shopify stores. Recently, more studies stated and insisted that there are many chances to enhance the impacts despite criticisms concerning anonymity security measures for Bitcoin transactions [8]. Moreover, the present architecture needs to be adapted to support the demand evolution of Bitcoin and improve in the research fields of data security and cryptography [9].

The exchange rate of Bitcoin volatility is difficult for common users and traders. Henceforth, a model needs to be idealized to describe the price behaviour of Bitcoin [10], which makes the unsettled helpful market. The author suggests that Bitcoin's high volatility is not the factor that generates it as a currency, as mentioned by [11]. However, the risk is reduced by [12] seeking solutions for ordinary users and traders. Then, predicting the price direction of assets is a practical task. An accurate prediction of price direction drives the trade and creates good returns for investors. The exchange rate of Bitcoin is maximized [13], which is described in the number of studies regarding time series, but these are currently relative. Many authors try to find the factors or attributes that provide more correlation and the price variation of Bitcoin [14]. The

behaviour of the exchange rate of Bitcoin is predicted by a few existing kinds of research [15]. One of the major problems with various Bitcoin predictions is the lack of sufficient analytical support to back up their claims. Some of the existing prediction models that fail to get superior outcomes are the Bayesian network model, Support Vector Regression, Linear Regression, and standard LSTM.

The objective of the paper is to select suitable attributes and use Deep Learning approaches in predicting the exchange rate of Bitcoin to acquire a higher accuracy than the current research. This research intends to establish a better trade-off in the Bitcoin prediction process. The simulation is done in the MATLAB 2020a environment with the advanced toolbox. The significant contributions of the proposed Bitcoin prediction approach rely on the following points:

Initially, an available online dataset like Yahoo or Coin Desk is considered to initiate the prediction process.

The input dataset needs to undergo some essential pre-processing steps to remove the missing data. Then, Principal Component Analysis (PCA) is applied to predict the most influential features from the pre-processed dataset followed by Technical Indicator extraction.

The most influencing Technical Indicators (T.I.) from the extracted Technical Indicators are examined using the most-adopted feature selection approaches like LASSO and Fisher Score. The essential features are analyzed, compared, and provided input to the network modelling phase.

The samples need to be partitioned as training and testing samples for learning the input data. The prediction is done with multiplicative LSTM, where the Cuckoo Search Optimizer with Game Theory Concept is incorporated with the proposed network model to perform prediction. The primary research novelty relies on this proposed idea.

Finally, various statistical measures are used for evaluation and comparative result highlights the significance of the proposed model over the existing ones.

The remaining paper is structured as follows: Section II covers the related works followed by the problem statement in Section III. Section IV covers the proposed methodology. Inferences, as well as experimental results of the proposed methodology, are explained in Section V. Finally, conclusions followed by future enhancements are discussed in Section VI.

2. Related works

The study generally comprised the empirical work for analyzing the determinants when the Bitcoin price issue is considered. First, Unal et al., [16] suggested forming the Bitcoin market with experimental traders without fundamentals and determining the interconnection between the search volume of Google Trends and Wikipedia and Bitcoin. The outcomes depict the relationship between the search queries and the price of Bitcoin that announces asymmetry that proposes the experimentation and helps to chase the trend that drives the dynamics of Bitcoin price in the cryptocurrency market.

The analysis of wavelet coherence is used by Hunag et al. [17] to look for the capable Bitcoin price drivers that have the source of fundamental, experimental, technical drivers, and peculiar properties of Bitcoin and are announced as both the practical one standard assets financially. The factors need to be identified by examining the coins with the help of cross-sectional empirical data by Neely et al. [18] to drive the values of cryptocurrencies in the technical region and consider the three essential drivers. They are (i) the cumulative computational power utilized in the process of mining for the cryptocurrency units, (ii) the unit production rate, and (iii) the cryptologic algorithm utilized for the protocol. The time series analytical mechanism needs to be executed with Bitcoin data daily.

The empirical analysis is designed by Hsu et al. [19] on Bitcoin's price determinants in the Chen et al. [20] model framework. The conventional determinants and the particular attributes are mentioned in the results related to digital currency, the forces of the market, and attention to the investors that drive the price of Bitcoin. It is more influencing and disagreeing about the macro-level financial developments that are the Bitcoin drivers for the lengthy duration. Moreover, sentiment is considered an irrational attribute that recommends the empirical market research of Bitcoin [21]. The sentiment is determined by Xie et al. [22] which is considered the deciding factor of Bitcoin volatility to identify the explanatory power of sentiment that contains a positive relationship with volatility, especially while the trades of Bitcoin are in the stage of a bubble. The user comments are examined in [23] by the communities of cryptocurrency online for Ethereum, Bitcoin, and Ripple for the price fluctuations predictions and the number of transactions.

The Granger causality model is implemented and attains the crawling data with 79.57% accuracy and the association between the price fluctuations. The number of transactions is considered with optimistic topics, replies, and comments. The dynamic interactions between the value of Bitcoin and the information system and literature on finance related to social media are analyzed by Chen et al. [24]. Social media is considered the essential predictor of the price of Bitcoin that views the 5% of active users who provide the many feasible messages having majority hypothesis.

Machine Learning (ML) is a type of Artificial Intelligence (AI) that can identify past and future data. It has various benefits over the prediction model where the outputs are the same as the appropriate outcomes; however, it enhances the prediction accuracy. The samples of ML approaches include SVM, NN, RF, and so on. The investigators include cryptocurrency in a portfolio to enhance model efficiency. Initially, SD is reduced and the successive outcomes offer investors huge allocation options. The cryptocurrency allocation is reported and the value ranges from 10% to 20% based on the risk tolerance of investors. The investigators also concentrate on time series data forecasting and ML algorithms like stochastic gradient boosting and RF are used and the outcomes demonstrate that the ML ensemble technique is adopted to acquire bitcoin values [25].

The empirical research does not implement the benefits to emerge the approaches with Machine Learning algorithms for the prediction of the price of Bitcoin and performing the analysis in the region is inadequate. The "latent source model" is used by Livieris et al. [25] to design the Bayesian regression that develops the binary classification to leverage and predict price variations of Bitcoin. The relationship between the price of Bitcoin and the price determinants is analyzed by Dutta et al. [26] using the Support Vector Machine algorithm that has variables, and attributes related to technical, and sentiment.

The Machine Learning algorithms are executed by Jang et al. [27] for examining the deciding factors of network features on the price of Bitcoin, and 55% accuracy is achieved. The Machine Learning algorithms are used to predict Bitcoin price by Khaiden et al. [28], having 98.7% accuracy related to daily price and 50% to 55% about high-frequency price. The accuracies of RNN (recurrent neural network), LSTM (long short-term memory), and the ARIMA (autoregressive integrated moving average) are compared by Zheng et al. [29] to predict the price of Bitcoin and reveal that LSTM achieves the most remarkable accuracy of about 52%.

The assumption related to other features is considered equal to that, as suggested by Occam's razor principle, and the simple model is utilized during the Machine Learning leverages. Arguments and shreds of evidence are considered firstly against the proposition. Occam's razor provides the same training set error is interpreted, and the

simple models need to be recommended due to the less generation error, which is wrong. The empirical analysis is proved with the interpretation that reveals the overfitting problem. It occurs due to the number of models rather than the complexity of models [30]. These are supported even though Occam's razor is contentious in modern science and generally in machine learning techniques. Queries are raised regarding whether the simple models need to be applied under prediction conditions and if the relatively simple statistical approaches can predict the price of Bitcoin with more accuracy. Practically, many prediction problems have actual results for implementing high VC-dimension models [30]. The two general paradigms are used. They are (i) complex models with some attributes and (ii) statistical methods having prominent attributes. Machine learning is extended by Occam's razor with the high volume of attributes and the high dimension datasets. New adaptive bounds are implemented and utilized for making the self-rounding machine learning algorithms. Occam's razor is considered for selecting feature subsets for releasing datasets with high dimensions. This method performs computational searches organized through the essential and fundamental ideas analyzed by the concept of whether the complications are inherent to score the functions based on the theory of Occam's razor for ligand affinity prediction from justifying the docking simulation. The insufficient choice of conventional machine learning issues is described to be formulated based on Occam's razor theory, and the proposed learning from the data is considered as an alternate source. Table 1 depicts the comparison of various existing approaches.

Table 1. Comparison of various existing approaches

References	Crypto-currency	Approaches	Prediction Outcomes
Saad et al., [2]	LTC and BTC	Multi-linear regression model	Recall → 44% (LTC) Recall → 59% (BTC)
Rebane et al., [3]	BTC	Linear Discriminant Analysis and Logistic regression	RMSE → 302 (ARIMA) RMSE → 603 (LSTM) RMSE → 381 (GRU)
Wu et al., [9]	BTC	GRU, LSTM and ARIMA	RMSE → 302, LSTM → 603, GRU → 381
Sherratt et al., [10]	BTC	Bi-LSTM, GRU and LSTM	LSTM MAPE → 1.12% RMSE → 410.3 Bi-LSTM MAPE → 5.9% RMSE → 2927

Le et al., [11]	ETH	Bi-LSTM, GRU and LSTM	LSTM MAPE→ 1.5% RMSE→ 59.5 Bi-LSTM MAPE → 0.2% RMSE→ 0.8
Kwak et al., [12]	LTC	Bi-LSTM, GRU and LSTM	LSTM MAPE→ 0.84% RMSE→ 3.06 Bi-LSTM MAPE →2.3% RMSE→ 4.3

3. Problem Statement

The investment process is completely based on the historical Bitcoin price. One of the essential strategies that the investors depend on is the construction of a Deep Learning network model to deal with the memory of prior values. Based on the prior value, the proposed model needs to predict the bitcoin value which is estimated to offer a greater return when sold. Due to the significance of the investment process, many investigators depend on revenue, this work concentrates on accurate prediction of Bitcoin price using Deep Learning approaches to attain a higher prediction model and aims to help the investors. Firstly, feature engineering needs to be conducted on the price granularity of Bitcoin using learning approaches. Many previous works concentrate on higher prediction accuracy without concerning the feature dimensions. The improved prediction algorithm is developed to predict the price of Bitcoin which is simpler to understand, follows, and takes decisions by the traders.

4. Methodology

This section discusses the broader analysis of the proposed methodology for Bitcoin prediction using Deep Learning and optimization approaches. Fig 1 depicts the overall research flow of the proposed idea. The simulation for prediction is done in the MATLAB 2020a environment, and various metrics like error rate and prediction accuracy were evaluated. The analyses with the technical indicators are also made to enhance the quality of prediction accuracy.

4.1. Data acquisition

To experiment with the above prediction model, 2 different datasets were collected. The first dataset was taken from Yahoo Finance with seven and half years of the daily price of Bitcoin. A total number of 2769 rows were collected which ranges from 17th September 2014 to 16th April 2022 which contains 7 features namely date, opening price, high price, low price, closing price, adjusted closing price, and transaction volume [31]. To evaluate the model one more dataset from the Coindesk was collected with 2722 observations ranging from 3rd November 2014 to 16th April 2022 [32]. This dataset consists of 5 features namely date, opening price, closing price, highest price, and lowest price for the 24 hrs. The dataset is split into 80-20 combinations of which 80% were taken for training and the remaining 20% were reserved for validation.

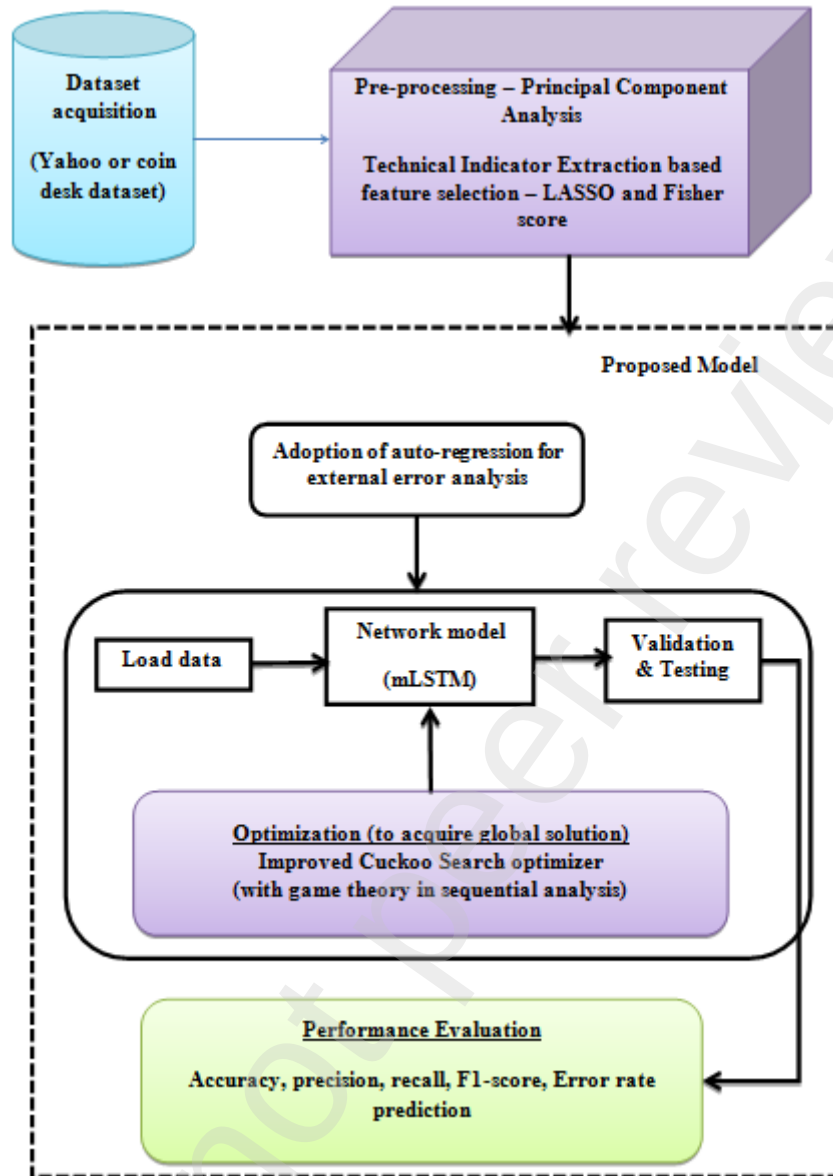


Figure 1. Flow Diagram of the proposed model

4.2. Technical Indicators

In this paper around 144 Technical Indicators were extracted using a predefined Python wrapper package provided by the TA-Lib [34]. Here, a few sample Technical Indicators are defined along with formula and variable descriptions. The following Table 2 is the summary of the technical indicators:

Table 2. Various Technical Indicators

Indicators	Formula
Simple moving average (10 days) [35]	$\frac{C_t + C_{t-1} + \dots + C_{t-10}}{10}$
Weighted moving average (10 days) [35]	$\frac{((n)) * C_t(n-1) * C_{t-1} + \dots + C_{10}}{(n + (n-1) + \dots + 1)}$

Stochastic K% [36]	$\frac{C_t - LL_{C-n}}{HH_{t-n} - LL_{C-n}} * 100$
Stochastic D% [36]	$\sum_{i=0}^{n-1} \frac{K_{t-i}\%}{n}$
Momentum [37]	$C_t - C_{t-n}$
Relative Strength Index (RSI) [37]	$100 - \frac{100}{1 + (\sum_{i=0}^{n-1} Up_{t-i}/n) / (\sum_{i=0}^{n-1} DW_{t-i}/n)}$
Moving average convergence divergence (MACD) [38]	$MACD(n)_{t-1} + \frac{2}{n} + 1 * (Diff_t - MACD(n)_{t-1})$
Larry William's R% [39]	$\frac{H_n - C_t}{H_t - L_n} * 100$
Accumulation/Distribution Oscillator (A/D) [40]	$\frac{H_t - C_{t-1}}{H_t - L_t}$
Commodity Channel Index (CCI) [41]	$\frac{M_t - SM_t}{0.015D_t}$
On-Balance-Volume (OBV) [42]	Current OBV = yesterday OBV's + today's trading volume Current OBV = Yesterday's OBV - Today's trading volume Yesterday's OBV = Today's OBV
Accumulation/Distribution Line [40]	$MFM = \frac{[(C - L) - (H - C)]}{H - L}$ MFV = volume for specific period * MFM
Average Directional Index (ADX) [37]	+DI -DI
Aroon Indicators [43]	$AroonUp \text{ line} \rightarrow [(\text{number of periods}) - (\text{number of periods since the highest high})] / (\text{number of periods}) * 100$

	AroonDown line→[(number of periods) – (number of periods since the lowest low)] / (number of periods)] * 100
Bollinger bands [44]	Middle line: 20-day simple moving average (SMA) Upper band: 20 – day SMA + (20 days SD * 2) Lower band: 20 – day SMA – (20 days SD * 2)
Ichimoku Clouds [45]	Specifies support level and resistance level
Fibonacci Indicators [46]	Swing high Swing low

4.3. Principle Component Analysis

Consider the dataset 'X' with 'm' rows specifying variables and 'n' columns specifying the Bitcoin prediction (observations) with a matrix of order 'm' row vectors and 'n' length as expressed in Eq. (1) [47]:

$$X = \begin{bmatrix} x_{1,1} & x_{1,2} & \dots x_{1,n} \\ x_{2,1} & x_{2,2} & \dots x_{2,n} \\ \vdots & \vdots & \vdots \\ x_{m,1} & x_{m,2} & \dots x_{m,n} \end{bmatrix} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_m \end{bmatrix} \quad (1)$$

The model requires linear transformation matrix X into $Y_{m \times n}$ using $P_{m \times m}$ with row vectors $p_1, p_2, \dots, p_m$ and column vectors $x_1, x_2, \dots, x_n$ as expressed in Eq. (2) [47]:

$$Y_{m \times n} = P_{m \times m} X = \begin{bmatrix} p_1 \cdot x_1 & p_1 \cdot x_2 & \dots p_1 \cdot x_n \\ p_2 \cdot x_1 & p_2 \cdot x_2 & \dots p_2 \cdot x_n \\ \dots & \ddots & \dots \\ p_m \cdot x_1 & p_m \cdot x_2 & \dots p_m \cdot x_n \end{bmatrix} \quad (2)$$

Here, $p_i \cdot x_j$ specifies the original dataset X projection over column P. Here, P rows represent principal component directions. PCA considers data variance and seeks original data de-correlation by predicting directions with maximal variance and uses direction to indicate the new flow of research. The initial phase of PCA analysis is subtracting the mean values from the dataset to perform normalization and weighted analysis. It is expressed as in Eq. (3) [47]:

$$X_{i,j} = Z_{i,j} - \bar{X}_j \quad (3)$$

Here, $X_{i,j}$ specifies a standardized dataset, $Z_{i,j}$ is standard data variables 'j' in sample i, and $\bar{X}_j$ specify sample means for variables j. The variable covariance is considered the measure of correlation, and all the probable covariance pairs among variables and diagonals are considered all the variables. It is expressed as in Eq. (4) [47]:

$$C_x = \frac{1}{n-1} XX^T = \frac{1}{n-1} \begin{bmatrix} x_1 x_1^T & x_1 x_2^T & \dots x_1 x_m^T \\ x_2 x_1^T & x_2 x_2^T & \dots x_2 x_m^T \\ \vdots & \ddots & \vdots \\ x_m x_1^T & x_m x_2^T & \dots x_m x_m^T \end{bmatrix} \quad (4)$$

Generally, significant variances are essential as they are related to some system dynamics, while smaller values specify the system noise. Thus, covariance C_y of transformed matrix $Y_{m \times n}$ needs to fulfil some preliminary requirements:

- 1) The signals are maximized by maximizing the diagonal entries, and
- 2) reduce the covariance among the variables by reducing the off-diagonal entries. The requirements mentioned above are fulfilled by selecting the transformation matrix

$P_{m \times m}$ with vectors $p_1, p_2, \dots, p_m$ as C_Y is like the diagonal matrix as shown in Eq. (5) [47]:

$$C_Y = \frac{1}{n-1} YY^T = \frac{1}{n-1} (PX)(PX)^T = \frac{1}{n-1} (PX)(P^T X^T) = \frac{1}{n-1} P(XX^T)P^T = \frac{1}{n-1} P P^T (XX^T) P^T \quad (5)$$

Here, $(XX^T)^T = XX^T$ and the squared symmetric matrix is diagonalized where the symmetric $S_{m \times m}$ is expressed as in Eq. (6) [47]:

$$S = EDE^{-T} \quad (6)$$

Here, E is an $m \times m$ matrix whose columns are eigenvectors of S , and D is a matrix that possesses eigenvalues of S as diagonal entries. The row matrix is eigenvectors given as $P = E^T$. Therefore, the covariance matrix of γ as in Eq. (7) [47]:

$$C_Y = \frac{1}{n-1} PSP^T = \frac{1}{n-1} E^T (EDE^T) E = \frac{1}{n-1} D; \text{ where } EE^T = I; I \rightarrow m \times m \quad (7)$$

4.4. Identity matrix

The covariance matrix is re-written as the summation of ℓ eigen values multiplied by the product of eigenvectors with transpose. Every principle component is considered the linear combination of provided x variables with coefficients and fulfils the following requirements. They are 1) maximizing Y variance; 2) The sum of squared coefficients of provided eigenvectors is considered 1; alike of norm value; and 3) some new principal components are uncorrelated with previously determined principal components. Consider, $\lambda_1, \dots, \lambda_t$ are eigenvalues of the covariance matrix C_Y with $\lambda_1 > \lambda_2 > \dots > \lambda_t$. Therefore, γ_i specifies the ratio of the total variance, and it is expressed as in Eq. (8) [47]:

$$\frac{\lambda_i}{\lambda_1 + \lambda_1 + \dots + \lambda_t} \quad (8)$$

The variation proportion of the k^{th} the principal component should be more significant or higher to eliminate information loss and be expressed as in Eq. (9) [47]:

$$\frac{\lambda_1 + \lambda_2 + \dots + \lambda_k}{\lambda_1 + \lambda_2 + \dots + \lambda_t} = 1 \quad (9)$$

The steps to perform PCA-based dimensionality reduction are given below. The following is the procedure to reduce the dimensionality of the d – dimensional dataset and highlight the k^{th} dimensional sub-space where $k < d$ with higher computational efficiency and maintains the information.

- Step 1: Standardization of token dataset and covariance matrix calculation.
- Step 2: Eigen values and Eigen vectors to be calculated and sorted.
- Step 3: Select n eigenvalues and vectors to reform the matrix.

4.5. Attribute selection with LASSO

The remarkable variable provides the essential scores that high dimensional LASSO amplifies by taking the absolute coefficients; hence, the variables' global oracle property is concerned entirely. The predictor's coefficient determination differs based on other predictors taken in this model due to the coefficient of regression explaining the predictor's impact with another predictor. Then, the predictor coefficient needs to have various values or the opposite sign with other predictors estimating in different linear models. In particular, multicollinearity having different signs frequently creates coefficient estimation with various signs across bootstrap samples. For example, when

the linear model has x_1 , x_2 , and x_3 , the x_1 and x_2 is significantly correlated with opposite signs. If $q_1 = 2$, then the primary bootstrap sample has the two variables of x_1 and x_2 and the secondary bootstrap sample has x_2 and x_3 . Therefore, the bootstrap samples are considered using the regression models that are written in Eq. (10) and Eq. (11) [48]:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon \quad (10)$$

$$y = \beta_0 + \beta_2 x_2 + \beta_3 x_3 + \epsilon \quad (11)$$

These bootstrap samples disperse the x_2 variable for the estimation of β_2 coefficient that is varied in two models. The mentioned models need to be depicted as follows in Eq. (12) and Eq. (13) [48]:

$$y = \beta_0 + \beta'_1 x_1 + \beta_2 x_2 + \epsilon \quad (12)$$

$$y = \beta_0 + \beta'_2 x_2 + \beta'_3 x_3 + \epsilon \quad (13)$$

However, the estimation of elastic-net attains a highly correlated. The variable has a similar sign, and only one significant variable is picked by LASSO when another one chooses the zero. Henceforth, the sum of the estimated coefficient of bootstrap samples minimizes the important score by considering the absolute value. Subsequently, the importance score is computed by LASSO by considering the sum of the absolute coefficient estimation of the bootstrap sample. If the $k_j^{(l)}$ represents the group of bootstrap samples indices in the j th procedure ($l=1$ or 2), and $|k_j^{(l)}|$ be the size of $k_j^{(l)}$ that has the j^{th} predictor. Therefore, the variables' importance score in LASSO is calculated through the average value of the absolute coefficient by eliminating the systematic bias using Eq. (14) [48].

$$I_j = \sum_{i \in k_j^{(1)}} b_{ij}^{(1)} / |k_j^{(1)}| \quad (14)$$

Thus, the final coefficients' estimate is calculated fully using Eq. (15) [48]:

$$\beta_j = \sum_{i \in k_j^{(2)}} b_{ij}^{(2)} / |k_j^{(2)}| \quad (15)$$

Consider the final coefficients' estimate that does not take the absolute coefficient values of the bootstrapping.

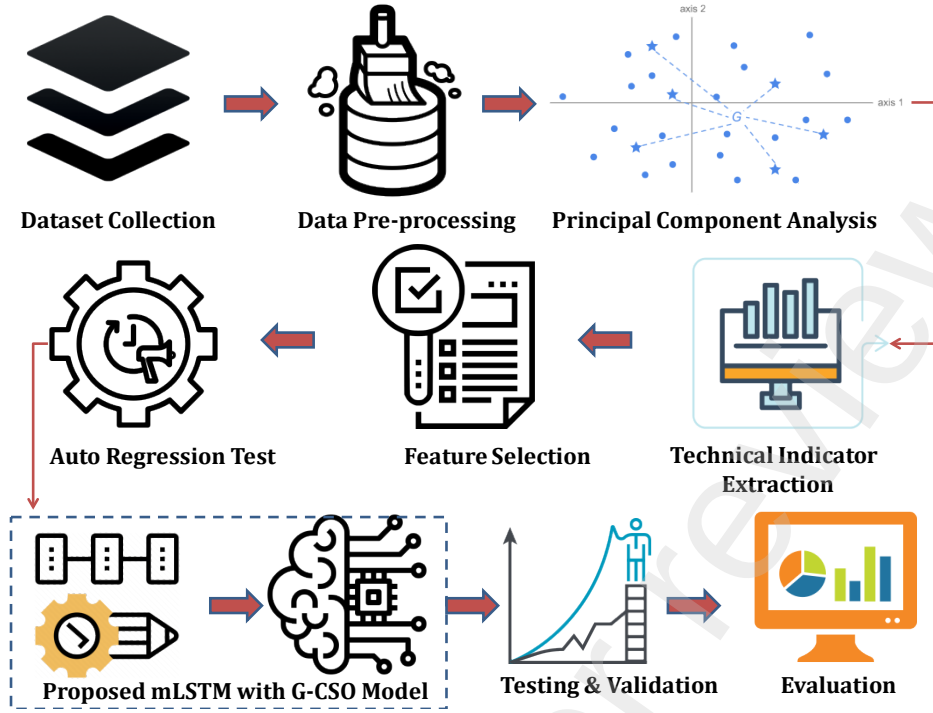


Figure 2. Generic view of proposed mLSTM – GCSO Model

4.6. Attribute selection with Fisher score

An important technique for reducing data with feature dimensions is the Fisher score. This research aims to identify the feature subset like the selected features spanned in the data space to increase the distance between data points in various classes when the distance between the data points in a similar class is minimized. Particularly, the training dataset is provided concerned to various classes c , the i th feature of Fisher score is calculated with Eq. (16) [49]:

$$FS(f_i) = \frac{S_b(f_i)}{\sum_{k=1}^c S_t^{(k)}(f_i)} \quad (16)$$

Here, $S_b(f_i) = \sum_{k=1}^c n_k (\mu_i^{(k)} - \mu_i)^2$ specifies the class scatter of i^{th} feature, n_k specifies the number of samples over the k^{th} class, $\mu_i^{(k)}$ defines the mean of i^{th} feature over the k^{th} class, μ_i specifies the mean of i^{th} feature over X , $S_t^{(k)}(f_i) = \sum_{j=1}^{n_k} (x_{ij}^{(k)} - \mu_i^{(k)})^2$ specifies the class scatter matrix of i^{th} feature w.r.t. k^{th} class, $x_{ij}^{(k)}$ defines the i^{th} feature value for the j^{th} sample in the k^{th} class.

4.7. Feature representation

This research adopts the various features attributed with a high dimension, which is the crucial insight explained before and the prediction of the price of Bitcoin using deep learning algorithms. The proper feature sets need to be selected daily for Bitcoin and predict the high-frequency prices through feature engineering. Feature selections are done by the following steps. Initially, the raw input data is pre-processed to remove the missing data. Then, PCA is applied to the pre-processed data to retrieve the most influencing features. With the above-retrieved features, Technical Indicator extraction is to be done. Then attribute selection technique (LASSO and Fisher Score) is applied, from the above step, the most important technical indicators with higher values generated by the LASSO and Fisher Score are selected and fed those technical indicators as input to

the prediction model for price prediction. Table 3 summarises the feature selection attributes using LASSO and Fisher scores. Some features like simple MA, weighted MA, momentum, stochastic K%, stochastic D%, RSI, MACD, LW R%, A/D oscillator, CCI, Bollinger Bands, and On-balance volume are considered as essential features. The maximal, minimal, mean, and SD of the features are provided in this table.

Table 3. Statistics Summary of feature selection attributes using LASSO and Fisher score.

Indicators	Maximum	Minimum	Mean	Standard Deviation
Simple MA	57156.84	950.11	17866.85	14718.28
Weighted MA	57451.35	960.88	17896.54	14740.62
Momentum	160.31	60.50	102.95	9.66
Stochastic K%	99.35	6.90	56.80	24.74
Stochastic D%	98.31	11.80	56.83	19.35
RSI	97.26	14.45	54.50	13.10
MACD	2076.34	-2118.36	139.10	509.28
LW R%	0.00	-100.00	- 42.73	30.26
A/D oscillator	8E+011	1E+007	2E + 010	2E+011
CCI	289.22	-322.22	12.94	87.98
Bollinger bands	388.70	302.17	20.98	94.2
On-Balance-Vol ume (OBV)	332.81	2785	1558.9	1226.095

4.8. Auto-regression for error analysis

The auto-regression combines regression and moving average models to perform the time series stationary via differencing. Thus, the differences are not needed explicitly. The residual sum of square-based error is minimal for the regression model. Therefore, it is considered the best among the bitcoin variable dependency on error terms and lagged values. Thus, it is used to measure the mean square error. The prediction is based on the testing samples, i.e. low price, open price, close price, and high price. It is used for processing the data, i.e. (p, d, q) where $p \rightarrow$ order of auto-regression; $d \rightarrow$ degree of first difference and q is the order of moving average. However, it is complex to choose the appropriate p, d, q . This regression process is based on the prediction values, where $AR(1)$ is the first-order process of auto-regression where the successive values rely on the present value. Similarly, $AR(2)$ specifies the current value based on two preceding values. The bitcoin data is divided based on testing and training data with a ratio of 80:20, which is used to measure the log and variable differences. Now compute the percentage error with the predicted price and the actual price, i.e. real value taken for the computation, and the expected value is attained with the auto-regression model. The error percentage is expressed as in Eq. (17) [50]:

$$\text{percentage error} = \frac{\text{predicted price} - \text{actual price}}{\text{actual price}} * 100 \quad (17)$$

The mean percentage error is evaluated using the percentage error. The mean percentage is provided below in Eq. (18) [50]:

$$\text{Mean percentage error} = \frac{1}{n} * \text{percentage error} \quad (18)$$

Here, n specifies the total number of terms in series. The data is loaded from a CSV file over the data frame (train). Consider a sample based on dataset attributes as in Table 4:

Table 4. Sample Dataset Attributes

	Date	Open	High	Low	Close	Adj Close	Volume
0	17-09-2014	465.864	468.174	452.422	457.334	457.334	21056800
1	18-09-2014	456.86	456.86	413.104	424.44	424.44	34483200
2	19-09-2014	424.103	427.835	384.532	394.796	394.796	37919700
3	20-09-2014	394.673	423.296	389.883	408.904	408.904	36863600
4	21-09-2014	408.085	412.426	393.181	398.821	398.821	26580100

Auto-regression based predicted, expected, and error rate:

- 1) 2513.745190, 2564.060000, and error = 1.962310%
- 2) 2566.007269, 2601.460000, and error = 1.369625%
- 3) 2604.348630, 2601.980000, and error = 0.090650%
- 4) 2605.558977, 2608.570000, and error = 0.115050%
- 5) 2613.835794, 2519.660000, and error = 3.778830%

4.9. Network modelling with multiplicative LSTM (mLSTM)

The proposed mLSTM network model is constructed with multiple combinations of LSTM prediction outcomes to enhance the accuracy. The network model is used over the sequential data as input devoid of any generality loss where the data samples (bitcoin values) change over time (Refer Fig 2). The input involves the sequential length parameters (time steps) depicted with the representatives of the finite time window. Therefore, the sample sequence vector is represented based on the changes in the input vector over time. The architectural model of mLSTM is composed of memory blocks. It includes three controlling gates input, forgets, and output gate. It is provided to preserve the knowledge (previous Bitcoin values) with a certain self-connection that keeps the temporal network state while the gates manage the update and information flow. For time series-based Bitcoin value prediction with input x_t , LSTM updates the memory cell c_t and hidden state output h_t based on various computations, preserved at multiple time step t . The expression is provided below:

$$i_t = \sigma(W_{xi}x_t + W_{hi}x_{t-1} + b_i) \quad (19)$$

$$f_t = \sigma(W_{xf}x_t + W_{hf}x_{t-1} + b_f) \quad (20)$$

$$o_t = \sigma(W_{xo}x_t + W_{ho}x_{t-1} + b_o) \quad (21)$$

$$c_t = f_t \otimes c_{t-1} + i_t \otimes \phi(W_{xc}x_t + W_{hc}h_{t-1} + b_c) \quad (22)$$

$$h_t = o_t \otimes \phi(c_t) \quad (23)$$

The formula to calculate the values for the input, forget, and output gates as well as hidden and cell states are mentioned in Eq. (19 – 23) [51]. Based on all these equations, W_{mn} specifies the connection weight from gate m and n , the bias parameter b_n to learn, $m \in \{x, h\}$ and $n \in \{i, f, o, c\}$. Additionally, $\otimes$ specifies the element-wise multiplication, σ specifies the standard logistic sigmoid function, ϕ specifies tanh function $\sigma(x) = 1/(1 + e^{-x})$ and $\phi(x) = (e^x - e^{-x})/(e^x + e^{-x})$. The memory cells (input, forget, and output) are specified as i_t, f_t and o_t , while c_t specifies the internal state of the memory cell at t time. The hidden layer value at t specifies the vector h_t while h_{t-1} specifies the memory cell output at the previous time. The mLSTM is used to construct the gate component, which helps to learn prediction errors.

The novelty of mLSTM relies on its effectual mechanism for quicker modification of cells' memory content and its corresponding internal dynamics cooperatively. Thus, the mLSTM can learn the non-linear statistical dependencies of Bitcoin values compared to other models. In the proposed mLSTM, every LSTM network is considered the predictor model, and it is effective when there are dependable diversities among the samples. The mLSTM parameters are varied based on the sequential length to enhance the diversity of the individual LSTM network.

The underlying concept for evaluating the various sequence length parameters is to inject diversity during the individual LSTM model. It is beneficial for higher non-linear statistical dependencies using multi-sequence length values to create mLSTM with several memory cells that show complementary effects while learning the characteristics and internal dynamics of time series-based data. The varying sequence length can handle the non-stationaries and dynamics of real-time Bitcoin value prediction. The forecasting value based on time series is specified as $(y^{(1)}, ..., y^{(N)})$ with N observations. It is specified in Eq. (24) [51]:

$$\hat{y}^k = \sum_{m=1}^M w_m \hat{y}_m^k \text{ for all } k = 1, ..., N \quad (24)$$

Here, $\hat{y}_m^k$ specifies the predicted output (k^{th}) time step attained with the m^{th} LSTM model, and w_m determines the related weight. The feature weight is allocated to the mLSTM models' prediction outcome, i.e. $0 \leq w_m \leq 1$ and $\sum_{m=1}^M w_m = 1$. The essential factor that includes the prediction is based on the weight combination $w_m, m = 1, ..., M$. The underlying time series weight determination is provided adaptively. The details of weight determination are provided based on the regression error of individual estimators like zero mean and uncorrelated. The weighted average shows minimal variance when the weights are inversely proportional to estimator variance. The properties offer theoretical analysis of weight determination. The combined weights are provided as in Eq. (25):

$$w_m^{k+1} = w_m^{(k)} + \lambda \Delta w_m^{(k)} \text{ for all } m = 1, ..., M \quad (25)$$

Here, the value of λ is set as 0.3 where $\Delta w_m^{(k)}$ specifies the inverse prediction error based on mLSTM. It is expressed as in Eq. (26):

$$\Delta w_m^{(k)} = \frac{1/\varepsilon_m^k}{\frac{1}{\varepsilon_m^{(1)}} + \frac{1}{\varepsilon_m^{(2)}} + ... + \frac{1}{\varepsilon_m^k}} \quad (26)$$

Here, ε_m^k specifies the past prediction error evaluated with k^{th} time step as expressed in Eq. (27):

$$\varepsilon_m^k = \sum_{t=k-v+1}^k \gamma^{k-t} e_m^{(t)} \quad (27)$$

Here, $0 < \gamma \leq 1$, $1 \leq v \leq k$ and $e_m^{(t)}$ show the prediction error at time step t of mLSTM. Here, ε_m^k is evaluated using the sliding time window formed using the v prediction errors. The forgetting factors are devised using the influence of the older prediction error. The weighted updation is done with the intricate patterns of successive prediction data. It is beneficial for handling the non-stationary properties and eliminates complex optimization for predicting the adaptive weight. However, optimization is essential for evaluating global and local solutions. The weights are updated for time series $k = 1, \dots, T$, which leads to the final weight $w_m^{(T)}$ ($m = 1, \dots, M$). The weights are allocated to the mLSTM and evaluated with Eq. (28):

$$w_m = \frac{w_m^T}{\sum_{n=1}^M w_n^{(T)}} \text{ for } m = 1, \dots, M \quad (28)$$

The weight computation needs to fulfil constraints like $0 \leq w_m \leq 1$ and $\sum_{m=1}^M w_m = 1$.

4.10. Optimizing weighted output with Gamic-Cuckoo Search Optimization approach (GCSO)

The conventional Cuckoo Search Optimization (CSO) explains the concept with the number of available nest n with fixed value where the cuckoo provides only one egg at t time and chooses the nest for random hatching the egg. The optimal nest is chosen for successive generations with the randomly chosen net, and others are updated via Levy's flight concept. Moreover, the host will discard the nest and set up a new one to predict the foreign bodies. The host bird probability discovers the foreign body (bird), set as p_a . The nest is equal to the solution vector, and initially, the iteration process is initiated with the solution vector, where the evaluation is done with the fitness function. The optimal solution is retained, and other solutions are updated based on the random walk through Levy's flight. Every solution vector is measured against the fitness function and contains the optimal solution. Discard the solution with probability P_a and produce the new solution based on the preference-based number of walks. Every solution is computed using the fitness function and maintains the optimal solution after one iteration. The random walk measurement with levy flight using the location and path of the nests are computed using Eq. (29) [52]:

$$x_i^{(t+1)} = x_i^{(t)} + \alpha \oplus Levy(\lambda) \quad (29)$$

Where $x_i^{(t)}$ specifies the i^{th} nest for the t^{th} generation, α specifies step control adopted to control the searching range. This search control value obeys the normal distribution. $\oplus$ Specifies multiplication and the $Levy(\lambda)$ specifies the random search path using Levy distribution. It is expressed as in Eq. (30) [52]:

$$Levy: u = t^{-\lambda}, (1 < \lambda \leq 3) \quad (30)$$

Subsequently, local and global search is modified local and global searches are modified during easier for a narrow search. It makes the algorithm less likely to fall

under optimal local value and triggers convergence. For random walk computation with preference, the nest location and path are updated using Eq. (31) [52]:

$$x_i^{(t+1)} = x_i^{(t)} + r(x_j^{(t)} - x_k^{(t)}) \quad (31)$$

Where $x_i^{(t)}$ specifies the i^{th} nest and t^{th} generation, $x_j^{(t)}, x_k^{(t)}$ are two nests in the t^{th} generation. The modification is done with the Min-Max analysis to map the consistent prediction of the Bitcoin rate. In the random walk iteration process through preference, the location updation with CSO depends on the overall evaluation and updation strategy. The entire nest is updated and evaluated using the fitness function. Specific nest dimensions degenerate in the overall updation process, leading to the nest being abandoned during the fitness function evaluation. It ignores the evolution of particular dimensions and wastes the number of function evaluations. The step length evaluation does not provide the complete use of solution information during the iteration process via the preference. Thus, the modified CSO adopts a weight updation and evaluation strategy using consistent prediction (gaming concept). This modification maintains the faster dimensionality convergence and increases the relationship with prediction consistency. It reduces the overall prediction timing. The group updation and evaluation process partitions the nest into various groups and adopts a random updation process like an overall updation and prediction strategy. The groups of sequentially updated with the sequential play of the gaming concept. Assume that the players choose their strategies simultaneously. The playing strategy is achieved consecutively. If the column player chooses strategy Q and the row player chooses the strategy P. The column player's role is to maximize the loss of row players (the game is zero-sum). Given P (worst case), an adversarial column player chooses Q for maximizing $M(P, Q)$, i.e., if the row player performs mixed strategies P where the payoff is expressed as in Eq. (32):

$$\max_Q M(P, Q) \quad (32)$$

It specifies the overall probability distribution of columns and $\min_P$ determines the general minimum probability distribution of rows. This extreme is due to the set of distributions over the finite space. The row players need to choose P to reduce Eq. (32). Therefore, the row player's loss is expressed as in Eq. (33):

$$\min_P \max_Q M(P, Q) \quad (33)$$

The mixed strategy P^* realizes this minimal value, called a min-max strategy. If the column player plays first and the row player selects its plat with the advantage of knowing the strategy of the column player. The loss of a row player is expressed as in Eq. (34):

$$\max_P \min_Q M(P, Q) \quad (34)$$

Where Q^* realizes the maximal is called a max-min strategy. The time complexity of the updating and evaluation strategy is provided as $g * T(f)$ where the time complexity with the updation strategy is $d * T(f)$ which diminishes the running time, g number of groups (players), d is the number of dimensions, $T(f)$ is running time of fitness function. The evolutionary dimensions of one player (row player or column

player) are ignored due to the degeneration of other available groups and their strategies. The proposed model uses the evolutionary information to help the current solutions to local search attain higher quality solutions (bitcoin prediction) and diminishes the number of iterations. The solutions (prediction rate) are constantly optimized, not wasting the updation opportunities. By grouping, the group parameters possess the same attributes, and the group dimensionality of the group adopts the random updation method, which intends to enhance the inter-dimensionality relationship. The group updation strategies are applied to the random walk component via the CSO preference algorithm. In particular real-time applications, the grouping conditions are different. The CSO algorithm optimizes the weight of the Bitcoin prediction values using Weighted Neural Networks (WNN). The weights of the connection among the hidden and input layers are represented as w_{ij} , the weighted connection among the hidden and output layers is defined as w_{jk} , the scaling, and translation factor is a_j and b_j specifies the basis function of various intermediate nodes, $i \in [1, m], j \in [1, h], k \in [1, p]$. The nest is the vector of the above parameters. The number of group dimensionality is expressed as in Eq. (35):

$$D = m * h + h + h + h * p \quad (35)$$

Where m is the number of neurons over the input layer, p is the number of neurons over the output layer, and h is the number of neurons over hidden layers. Thus, based on the various nest attributes, the h – dimensionality vector is a group, and the entire vector is partitioned into D/h groups. It is expressed as in Eq. (36):

$$[\omega_{11} \dots \omega_{\mu 1}, \omega_{12} \dots \omega_{\mu 2}, \dots, \omega_{1\eta} \dots \omega_{\mu \eta}, \omega_{11} \dots \omega_{\eta 1}, \omega_{12} \dots \omega_{\eta 2}, \dots, \omega_{1\pi} \dots \omega_{\eta \pi}, \alpha_1, \alpha_2, \dots, \alpha_\eta, \beta_1, \beta_2, \dots, \beta_\eta] \quad (36)$$

The pseudocode for the GCSO algorithm is expressed as below:

Pseudocode: GCSO algorithm	
Pre-process: setting up the nests $x_i (i = 1, 2, \dots, n), T_{max}, P_a$	
Compute the fitness function $f(x_i)$;	
Initialize the best solution x_{best} ;	
Update x_i using Levy's flight;	
For $i = 1$	
While $i < n$ do	
Generate the random number (sequential manner using gaming strategies) r ;	
If $r > P_a$ then	
For $j = 1; h:d - h + 1$;	
$x_c = x_i$;	
$x_c[j:j + h + 1] = x_c[j:j + h + 1] + r(x(k, j:j + h - 1) - x[i, j:j + h - 1]);$	
If $x_c < x_i$ then //selection of random groups (row and column-wise players)	
$x_i = x_{best}$ //weighted computation from the output of mLSTM	
End	
End	
$i = i + 1$;	
End	
Output: x_{best}	//bitcoin prediction

5. Result and Discussions

5.1. Evaluation indicator

This section is not mandatory but can be added to the manuscript if the discussion is unusually long or complex. The simulation of the proposed model is done in the MATLAB 2020a environment. The confusion matrix needs to be constructed from the mentioned models, and the results must be divided into four divisions. The statistics in the confusion matrix regarding the classification of accurate data and the seven classifier algorithms generate the predictions. Every method's performance needs to be assessed from the results in the matrix. The confusion matrix has rows and columns that reveal the real instances and the classes that need to be predicted. The outcome is true positive when the real label is 1, and the prediction is also 1. The result is true negative when the real label is 0, and the prediction is 0. The outcome is false positive when the real label is 0, and the prediction is 1. The result is a false negative when the real label is 1 and the prediction is 0. The evaluation indicators of precision, accuracy, f1-score, and recall need to be computed using the below formulae based on the confusion matrix.

$$Accuracy = \frac{tp + tn}{tp + tn + fp + fn} \quad (37)$$

$$Precision = \frac{tp}{(tp + fp)} \quad (38)$$

$$Recall = \frac{tp}{(tp + fn)} \quad (39)$$

$$F1 - score = \frac{2 \times Precision \times Recall}{(Precision + Recall)} \quad (40)$$

The formula to calculate the values of accuracy, prediction, recall, and F1 – score is mentioned in Eq. (37 – 40) [53]. Here, tn represents the true negative, tp represents the true positive, fn represents the false negative, and fp represents the false positive. The confusion matrix represents the ratio between the false or true and negative or positive classifications, which is needed for the rating measures derivation. The total number of accurately classified predictions like price down, up, and no change is defined as accuracy. The specificity, sensitivity, and precision are examined to combat the inherent class imbalance, such as the increased price of Bitcoin. Sensitivity represents how good a model can be to detect the positive. Specificity means how good a model can be to avoid the wrong alarms. Lastly, how much positively classified prediction is suitable is defined by precision. The comparison and the evaluation of accuracy regression are determined by MSE (Mean Square Error), RMSE (Root Mean Square Error), MAPE (Mean Absolute Percentage Error), and MAE (Mean Absolute Error).

The MAE is used to measure the average of actual and predicted value difference but it doesn't resolve the correctness of the error. If the value of MAE is small, then the entrusted model is the best to project the future data. To calculate Mean Absolute Error using Eq. (41) [33].

$$\text{Mean Absolute Error} = \frac{1}{M} \sum_{k=1}^M |Z_k - Z_j| \quad (41)$$

where Z_k and Z_j stand for the actual output of model prediction respectively. 582

The MSE is a habitual regression metric for assessing the regression problem. The outcome of MSE will be a non-zero value that stipulates the average squared mean error of the predicted value with the original value. To calculate Mean Square Error using Eq. (42) [33]. 583 584 585 586

$$\text{Mean Squared Error} = \frac{1}{M} \sum_{k=1}^M (Z_k - Z_j)^2 \quad (42)$$

where Z_k and Z_j stand for the actual output of model prediction respectively. 588

The RMSE is homogeneous to MSE but RMSE is enumerated by putting in the square root of the MSE result. The below formula is used (Eq. (43)) to calculate the Root Mean Squared Error where Z_k and Z_j stand for the actual output of model prediction respectively [33]. 589 590 591 592

$$\text{Root Mean Squared Error} = \sqrt{\frac{1}{M} \sum_{k=1}^M (Z_k - Z_j)^2} \quad (43)$$

Mean Absolute Percentage Error is used to estimate the model's prediction accuracy. The given formula is used (Eq. (44)) to calculate the Mean Absolute Percentage Error where Z_k and Z_j stand for the actual output of model prediction respectively [33]. 594 595 596

$$\text{Mean Absolute Percentage Error} = \frac{1}{M} \sum_{k=1}^M \left| \frac{Z_k - Z_j}{Z_k} \right| \quad (44)$$

where Z_k and Z_j stand for the actual output of model prediction respectively. 598

5.2. Evaluation 599

The most excellent result is attained by mLSTM in terms of RMSE when compared to other standard LSTM, Bi-LSTM, and GRU models, as shown in Table 6 and the same plotted in Fig. 3 and Fig. 4. The price needs to be predicted, increasing slowly every day upon the prediction analysis of auto-regression. There is no false positive in the model. Because the class imbalance in the prediction area of auto-regression is a critical cause for the mentioned reason price needs to be continually enhanced. High specificity and high precision are achieved, and specificity is 91%. Ultimately better performance is not suggested and it performs considerable work to find the direction of price changes. All models are finding challenges to learning efficiently from the data that depend on the outcomes of validated data. The error rate is reduced to below 1% on training data by the model. LSTM achieves an error rate of about 80.7% on the validated data. The NN models achieve 50.25% and 52.78% accuracies considering the marginal enhancement across the odd one contained in binary classification such as price down versus price up, namely 50%. When using the temporal length across 50 days, the existing network model is not efficient. Contrarily, better performance is done by LSTM in the range of 50 to 100 in a day with 100 days that generate better performance. 600 601 602 603 604 605 606 607 608 609 610 611 612 613 614 615

Table 5. Execution time 616

Model	Epochs	CPU	GPU
mLSTM – GCSO	50	59.71s	38.85s

mLSTM – GCSO	500	1505s	888.34s
mLSTM – GCSO	1000	3001.69s	1746.87s

Table 5 compares the proposed model with a different number of epochs in terms of CPU and GPU executions. Similar batch sizes and 50 temporal lengths are selected for comparability for mLSTM. CPU performs well with the help of GPU significantly. GPU trains about 67.7% faster than CPU concerning overall training time. When the LSTM trained 70.7% faster on GPU, the mLSTM trained 58.8% faster on GPU. The CPU distributes the algorithm across seven threads from the monitored performance in Glances. The 384 CUDA cores are comprised of GPU that gives higher parallelism. Those models are small, and relatively concerning the data with two layers. The advantage of establishing the GPU is higher for deeper models with more layers or more extensive datasets.

Due to the increased number of activation functions, the LSTM takes a longer time to train than the traditional network model with the network parameters, and the number of equations is improved to perform with the help of LSTM. The queries of value have arisen with the use of LSTM because of the maximized calculation. Small margins generate all the differences in the prediction of the financial market. The usage of LSTM is proven as the outcome. The light enhancements are not justified in other areas concerning performance for the enhancement in the calculation.

This section depicts the results attained with the conventional LSTM, bi-directional LSTM, and Gated Recurrent Unit (GRU) with the proposed mLSTM. For the provided dataset, the outcomes are illustrated in Table 6. The model which provides lower MAPE and RMSE is considered the best model. Based on the observed value, all the other LSTM-based variants provide nominal outcomes compared to the GRU. The RMSE of the existing GRU is lesser, which has the competency in predicting the bitcoin rate-based dependencies compared to other approaches like conventional LSTM and bi-LSTM and not with the proposed mLSTM. It is due to the previous price's dependencies and depicts the comparison between the prediction and actual results. The simulation outcomes of these models specify the prediction outcomes. Table 6 shows the RMSE and MAPE comparison.

Table 6. RMSE and MAPE comparison

Models	RMSE	MAPE
LSTM	411.400	1.1243%
Bi-LSTM	2930.005	5.999%
GRU	175.130	0.2455%
mLSTM (proposed)	160.123	0.1260%

Table 7 depicts the comparison of mLSTM for diverse trading days using MSE. The comparison is made between two diverse datasets Yahoo and Coindesk. The performance of mLSTM for diverse trading days using MSE is analyzed for 2, 5, 10, 20, 30, 60, and 90 days. The MSE of Yahoo is 0.10015, 0.13345, 0.02589, 0.10030, 0.03645, 0.14505, and 0.12562; while the MSE of Coindesk is 0.10000, 0.14565, 0.17564, 0.02450, 0.01658, 0.13564 and 0.13258 (See Fig 5).

Table 7. Performance of mLSTM for various trading days using MSE

Dataset	2 days	5 days	10 days	20 days	30 days	60 days	90 days
Yahoo	0.10015	0.13345	0.02589	0.10030	0.03645	0.14505	0.12562
Coindesk	0.10000	0.14565	0.17564	0.02450	0.01658	0.13564	0.13258

Table 8 depicts the performance valuation of the Yahoo dataset with diverse approaches during the testing phase. Some approaches like GA-LSTM, PSO-LSTM, CSO-LSTM, MCSO-Att-mLSTM, and mLSTM - GCSO are used for comparison. The MSE of GA-LSTM is 0.59450, PSO-LSTM is 0.39130, CSO-LSTM is 0.22629, MCSO-Att-mLSTM is 0.21936 and mLSTM - GCSO is 0.11930. The RMSE of GA-LSTM is 0.76105, PSO-LSTM is 0.63550, CSO-LSTM is 0.46560, MCSO-Att-mLSTM is 0.47832 and mLSTM - GCSO is 0.01423 and MAE of GA-LSTM is 0.28756, PSO-LSTM is 0.18560, CSO-LSTM is 0.12315, MCSO-Att-mLSTM is 0.11968 and mLSTM - GCSO is 0.10056 (See Fig 6).

Table 8. Performance evaluation of the Yahoo dataset with various existing (benchmark) approaches during the testing phase

Dataset	MSE	RMSE	MAE
GA-LSTM	0.59450	0.76105	0.28756
PSO-LSTM	0.39130	0.63550	0.18560
CSO-LSTM	0.22629	0.46560	0.12315
MCSO-Att-mLSTM	0.21936	0.47832	0.11968
mLSTM - GCSO	0.11930	0.01423	0.10056

Table 9 depicts the performance evaluation of Coindesk with various existing (benchmark) approaches during the testing phase. The MSE of GA-LSTM is 0.64340, PSO-LSTM is 0.34540, CSO-LSTM is 0.26930, MCSO-Att-mLSTM is 0.14345 and mLSTM - GCSO is 0.10546. The RMSE of GA-LSTM is 0.80216, PSO-LSTM is 0.58776, CSO-LSTM is 0.51890, MCSO-Att-mLSTM is 0.37875 and mLSTM - GCSO is 0.01112. The MAE of GA-LSTM is 0.32170, PSO-LSTM is 0.17274, CSO-LSTM is 0.13460, MCSO-Att-mLSTM is 0.07174 and mLSTM - GCSO is 0.00850 (See Fig 7). Fig. 8 and Fig. 9 represent the accuracy and loss of the proposed Deep Learning model in terms of training and validation phases.

Table 9. Performance evaluation of the Coindesk dataset with various existing (benchmark) approaches during the testing phase

Dataset	MSE	RMSE	MAE
GA-LSTM	0.64340	0.80216	0.32170
PSO-LSTM	0.34540	0.58776	0.17274
CSO-LSTM	0.26930	0.51890	0.13460
MCSO-Att-mLSTM	0.14345	0.37875	0.07174
mLSTM - GCSO	0.10546	0.01112	0.00850

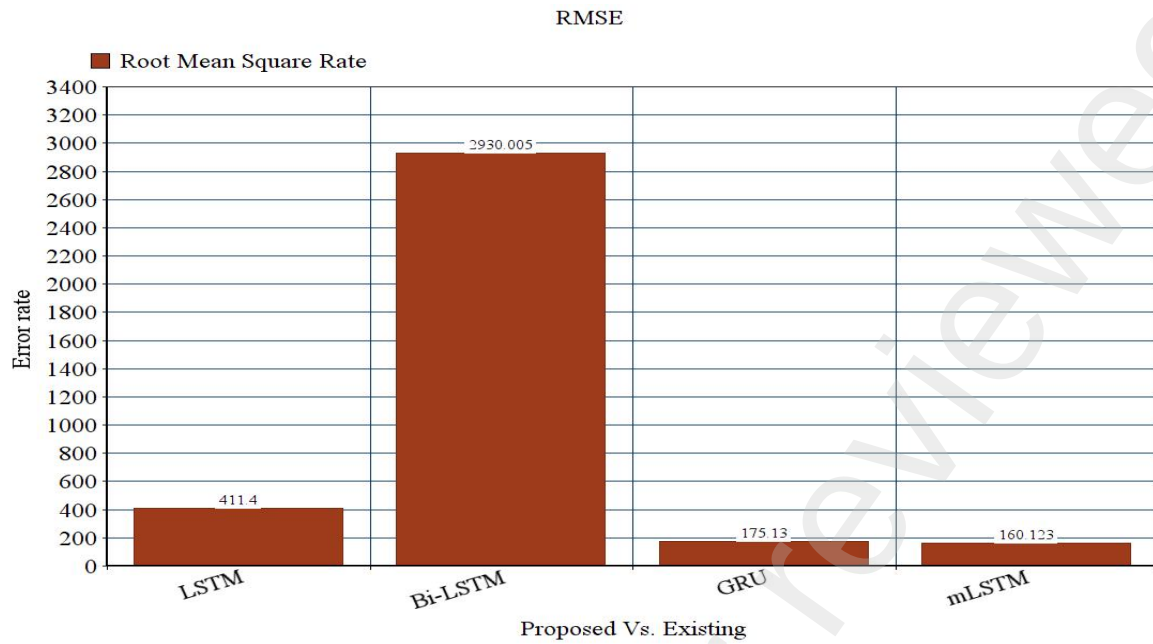


Figure 3. RMSE comparison

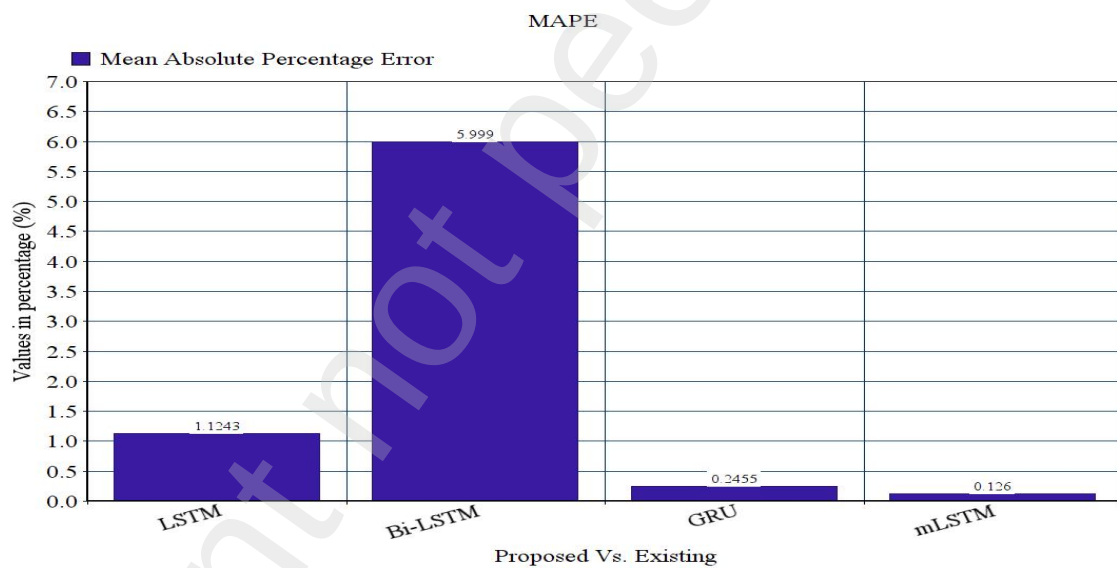


Figure 4. MAPE comparison

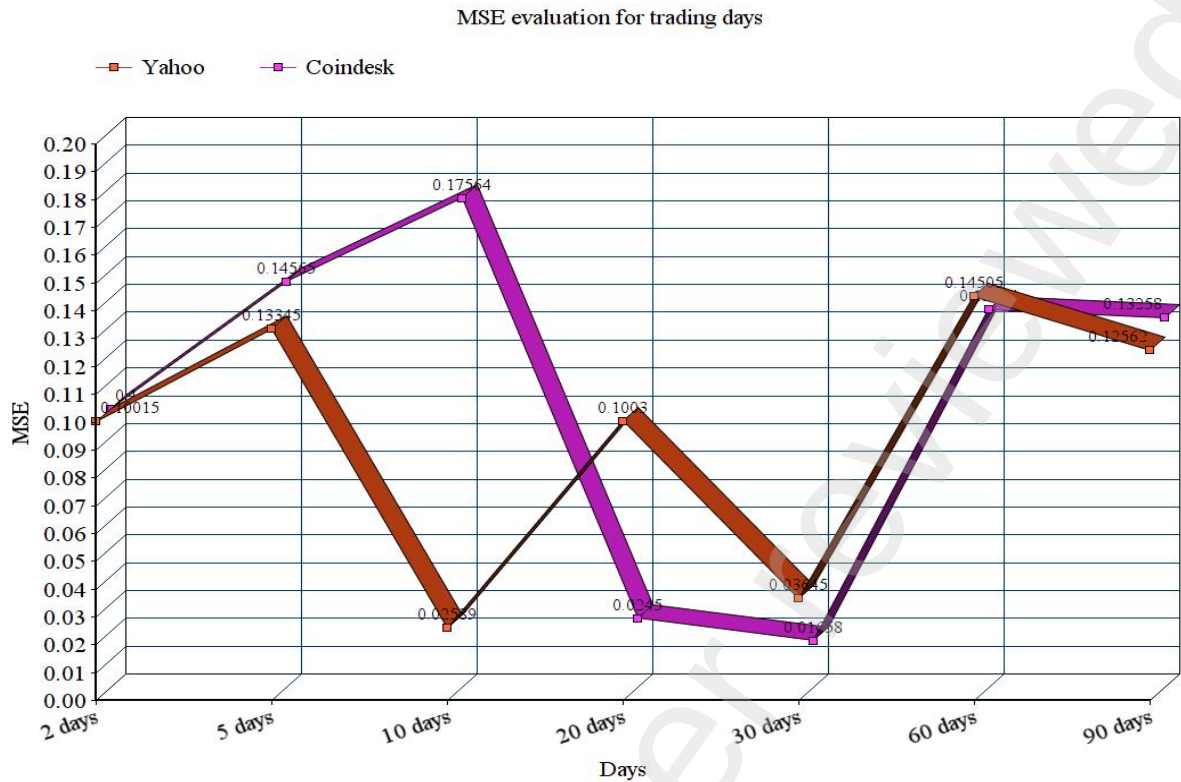


Figure 5. MSE evaluation for successive trading days

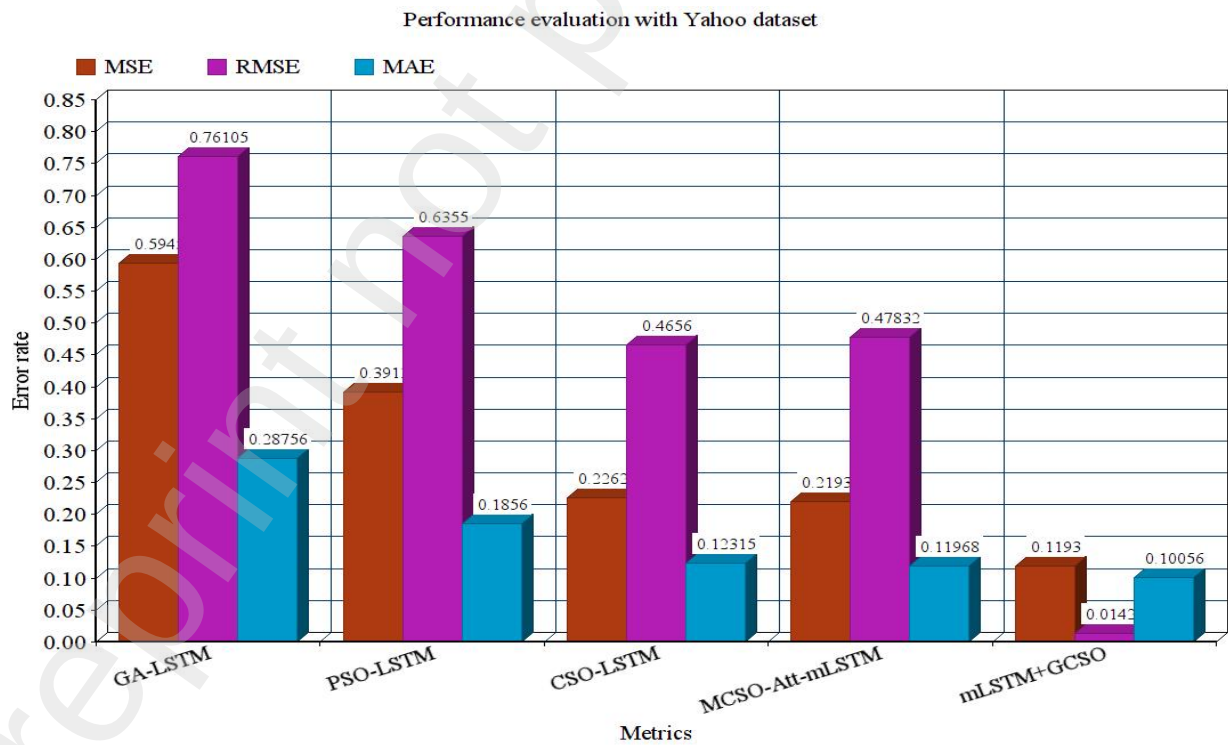


Figure 6. Performance evaluation with Yahoo dataset

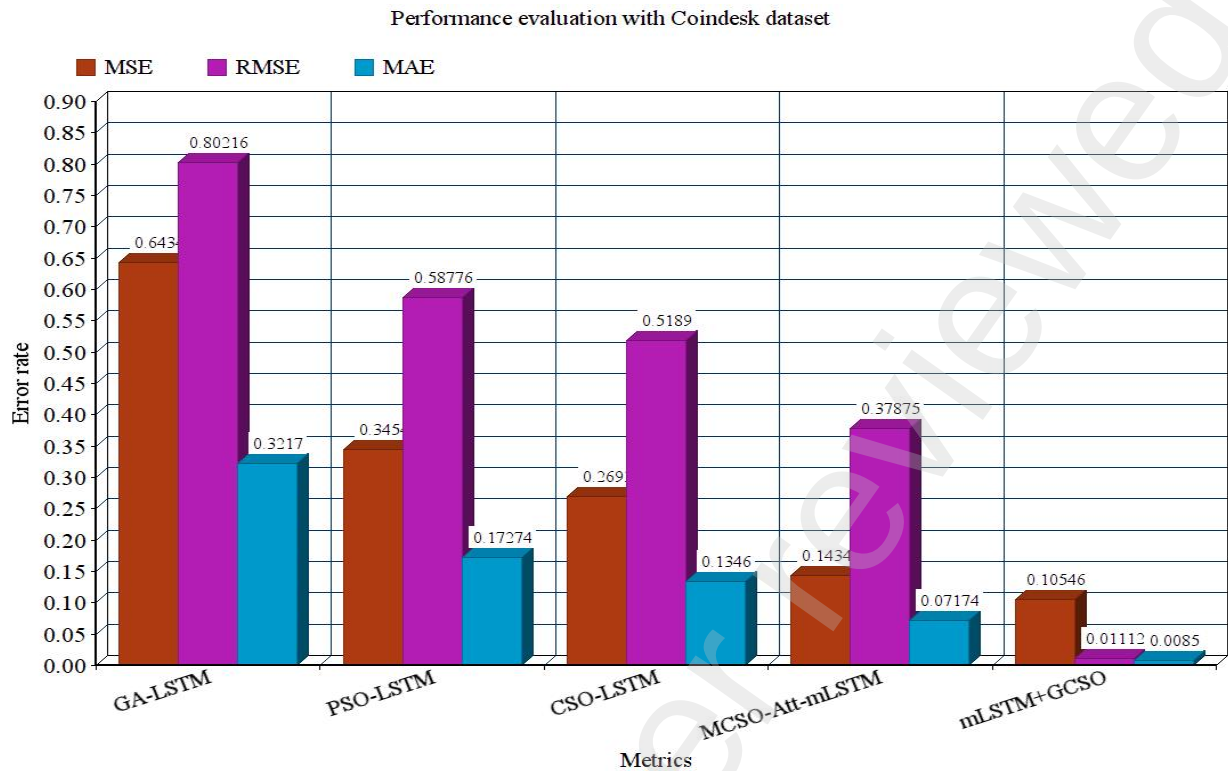


Figure 7. Performance evaluation with Coindesk dataset

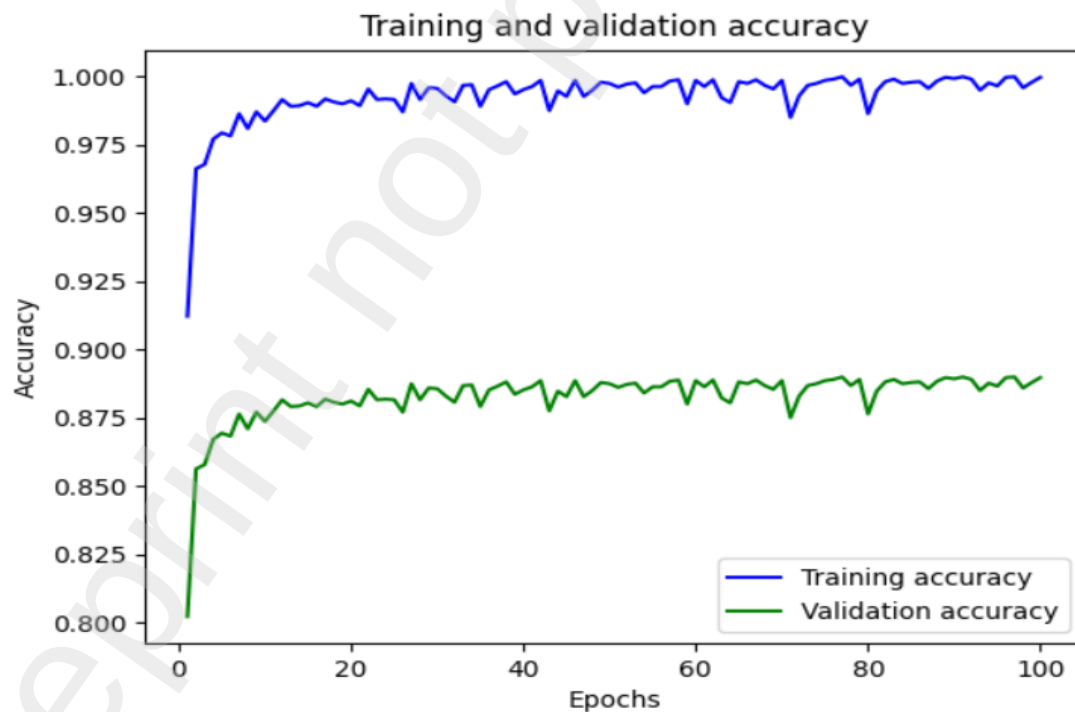


Figure 8. Training and validation accuracy

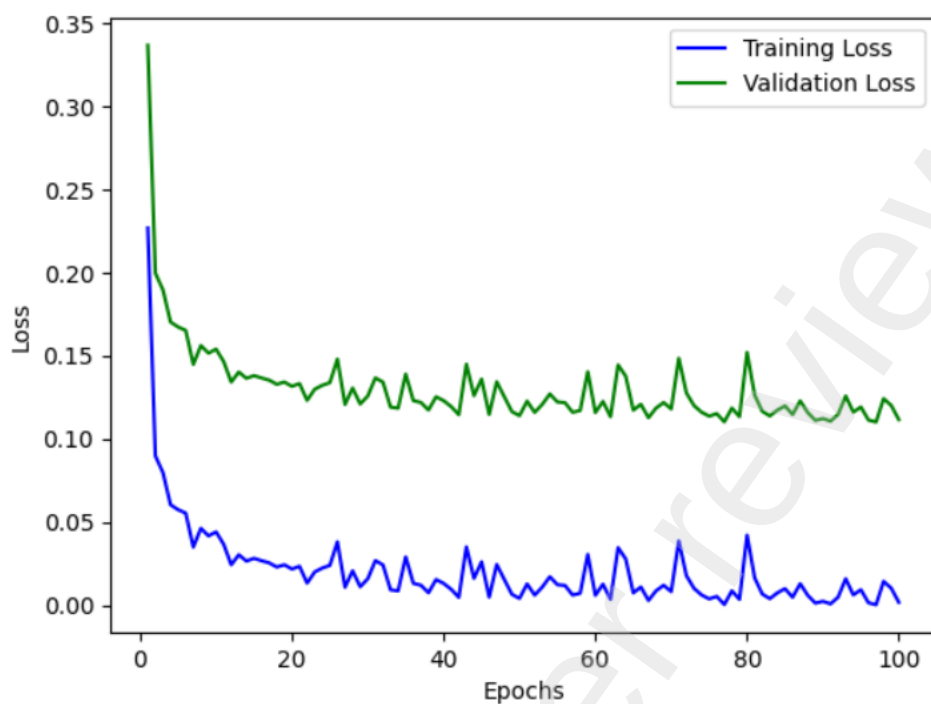


Figure 9. Training and validation loss

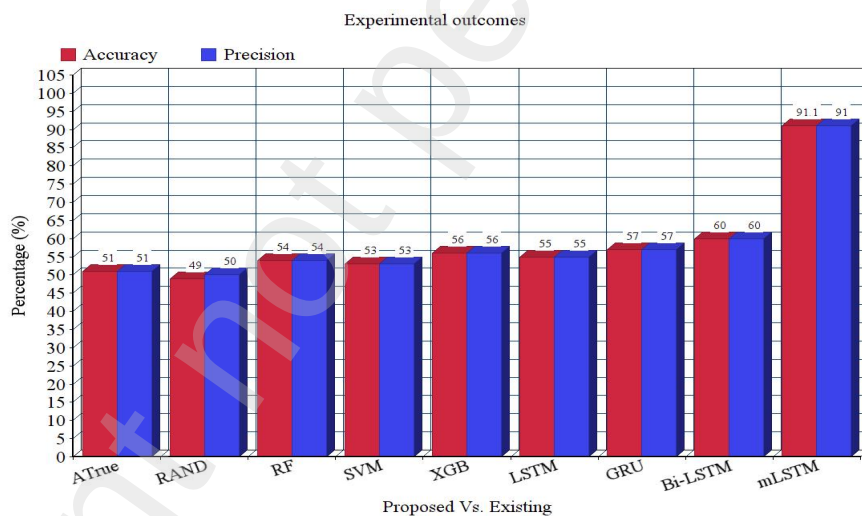


Figure 10a. Prediction outcomes (accuracy and precision)

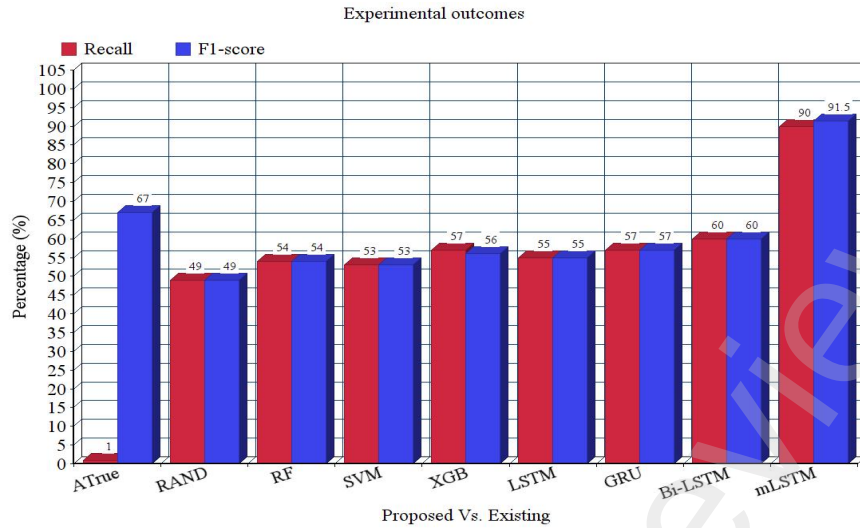
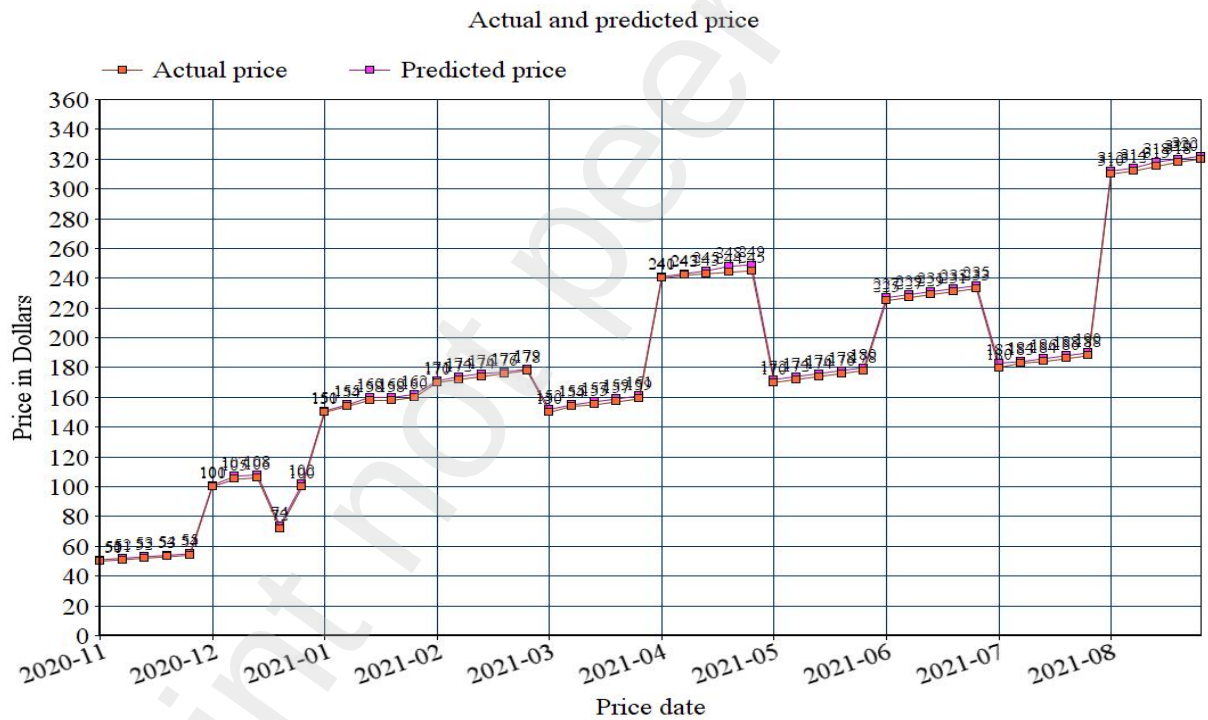


Figure 10b. Prediction outcomes (Recall and F1-score)



SVM	53	53	53	53
XGB	56	56	57	56
LSTM	55	55	55	55
GRU	57	57	57	57
Bi-LSTM	60	60	60	60
mLSTM - GCSO	91.1	91	90	91.5

Fig 10a and Fig 10b depict the comparison of proposed vs. existing approaches. The prediction accuracy of the anticipated model is 91.1% which is 40.1%, 42.1%, 37.1%, 38.1%, 35.1%, 36.1%, 34.1%, and 31.1% higher than other approaches like ATrue, RAND, RF, SVM, XGB, LSTM, GRU and Bi-LSTM. The precision of the proposed 91% which is 40%, 41%, 37%, 38%, 35%, 36%, 34%, and 31% higher than ATrue, RAND, RF, SVM, XGB, LSTM, GRU, and Bi-LSTM. The recall of the proposed model is 89% which is 89%, 41%, 36%, 37%, 33%, 35%, 33%, and 30% higher than ATrue, RAND, RF, SVM, XGB, LSTM, GRU and Bi-LSTM. The F1-score of the anticipated mLSTM is 91.5% which is 24.5%, 42.5%, 37.5%, 38.5%, 35.5%, 36.5%, 34.5%, and 31.5% higher than ATrue, RAND, RF, SVM, XGB, LSTM, GRU and Bi-LSTM (See Table 10). Based on this analysis (Fig 11), it is proven that the proposed mLSTM - GCSO works well for the prediction of bitcoin rate compared to other approaches.

6. Conclusion

The objective of the paper is to model Bitcoin price with an efficient learning approach known as mLSTM - GCSO model. Preliminary processes like pre-processing, feature extraction, and prediction are utilized for this task. Here, the pre-processing is performed to enhance the prediction quality by cleaning the data and the most-influencing technical indicators are considered for analysis where the extracted features are fed as an input to the modelling phase where the prediction is achieved with the proposed mLSTM - GCSO. The prediction accuracy is 91.1%, precision is 91%, recall is 90% and F1-score is 91.5% which is comparatively higher than other approaches like Atrue, RAND, RF, SVM, XGB, LSTM, GRU, and Bi-LSTM models. Performance metrics such as RMSE, MSE, and MAE are also evaluated and guide us in making informed decisions about the model we have proposed. The major research constraint is the lack of computational complexity analysis for the proposed model. However, it can be resolved in future work and some other deep learning-based prediction approaches can also be adopted to evaluate the shortcomings of the proposed model.

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